Exercise - 5.1

1. Expand (i) $\left(2x^2 - \frac{3}{x}\right)^3$ (ii) $\left(2x^2 - 3\sqrt{1-x^2}\right)^4 + \left(2x^2 + 3\sqrt{1-x^2}\right)^4$.
2. Compute (i) 102^4 (ii) 99^4 (iii) 9^7.
3. Using binomial theorem, indicate which of the following two number is larger: $(1.01)^{1000000}$, 10000.
4. Find the coefficient of x^{15} in $\left(x^2 + \frac{1}{x^3}\right)^{10}$.
5. Find the coefficient of x^2 and the coefficient of x^6 in $\left(x^2 - \frac{1}{x^3}\right)^6$.
6. Find the coefficient of x^4 in the expansion of $(1+x^3)^{50}(x^2 + \frac{1}{x})^5$.
7. Find the constant term of $\left(2x^3 - \frac{1}{3x^2}\right)^5$.
8. Find the last two digits of the number 3^{600}.
9. If n is a positive integer, using Binomial theorem, show that, $9^{n+1} - 8n - 9$ is always divisible by 64.
10. If n is an odd positive integer, prove that the coefficients of the middle terms in the expansion of $(x+y)^n$ are equal.
11. If n is a positive integer and r is a nonnegative integer, prove that the coefficients of x^r and x^{n-r} in the expansion of $(1+x)^n$ are equal.
12. If a and b are distinct integers, prove that $a - b$ is a factor of $a^n - b^n$, whenever n is a positive integer. [Hint: write $a^n = (a - b + b)^n$ and expand]
13. In the binomial expansion of $(a+b)^n$, if the coefficients of the 4th and 13th terms are equal then, find n.
14. If the binomial coefficients of three consecutive terms in the expansion of $(a+x)^n$ are in the ratio $1 : 7 : 42$, then find n.
15. In the binomial expansion of $(1+x)^n$, the coefficients of the 5th, 6th and 7th terms are in AP. Find all values of n.
16. Prove that $C_0^2 + C_1^2 + C_2^2 + \cdots + C_n^2 = \frac{(2n)!}{(n!)^2}$.

5.4 Finite Sequences

A sequence is a list of elements with a particular order. While the idea of a sequence of numbers, $a_1, a_2, \cdots$, is straight forward, it is useful to think of a sequence as a function whose domain is either the set of first n natural numbers or $\mathbb{N}$. Throughout this chapter, we consider only sequences of real numbers and we will refer to them as sequences. The arithmetic sequences and geometric sequences are also known as arithmetic progressions(AP) and geometric progressions (GP). Let us recall, the basic definitions of sequences and series.

- If X is any set and $n \in \mathbb{N}$, then any function $f : \{1, 2, 3, \ldots, n\} \to X$ is called a ***finite sequence*** on X and any function $g : \mathbb{N} \to X$ is called an ***infinite sequence*** on X. The value $f(n)$ of the function f at n is denoted by a_n and the sequence itself is denoted by (a_n).
- If the set X happens to be a set of real numbers, the sequence is called a numerical sequence or a sequence of real numbers.
- Though every sequence is a function, a function is not necessarily a sequence.
- Unlike sets, where elements are not repeated, the terms in a sequence may be repeated. In particular, a sequence in which all terms are same is called a constant sequence.
- A useful way to visualise a sequence (a_n) is to plot the graph of $\{(n, a_n) : n \in \mathbb{N}\}$ which gives some details about the sequence.